Regression

Logistic Regression

Classification

- **Regression.** The response variable is quantitative.
- **Classification.** The response variable is qualitative. Examples:
  - eye color $\in \{\text{blue}, \text{green}, \text{brown}\} := \mathcal{C}$;
  - email $\in \{\text{spam}, \text{nonspam}\} := \mathcal{C}$.
- **Number of Classes:** K.
- **Response** $Y \in \{1, ..., K\}$.
- **The goal** is to learn a function (*the classifier*) $C(X)$ that takes as input the feature vector X and tries to predict its class Y, i.e., $C(X) \in \mathcal{C}$.
- We are interested in estimating the probabilities that X belongs to each class in $\mathcal{C}$.

Classification

Let

$$p_k(x) = Pr(Y = k|X = x), k = 1, ..., K.$$

Then a very natural classifier (known as Bayes optimal classifier) is

$$C(x) = k \text{ if } p_k(x) = \max\{p_1(x), ..., p_K(x)\}.$$

- Estimate the probabilities $p_k(x)$, $k = 1, ..., K$;
- Assume we only have two classes, i.e., $K = 2$ and a single feature (the vector X is one-dimensional). Let

$$p(X) := Pr(Y = 1|X).$$

Will a linear regresion model $p(X) = \beta_0 + \beta_1 X$ be a good choice for estimating the true probability $p(X)$?

Logistic Regression

Let $K = 2$ and X be one-dimensional (scalar values).

- A linear probability model

$$p(X) = \beta_0 + \beta_1 X$$

  is not appropriate since $\beta_0 + \beta_1 X$ can be outside $[0, 1]$.

- **Logistic regression model**:

$$p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}.$$

- Can we get to a linear model? The log odds or logit transformation

$$\log\left(\frac{p(X)}{1 - p(X)}\right) = \beta_0 + \beta_1 X.$$

  (by log we mean the **natural log**: ln)

Logistic Regression. Estimating β_0 and β_1

Consider training data $\{(x_i, y_i) : i = 1, ..., n\}$ with $y_i \in \{0, 1\}$ for any value of i. Consider the likelihood function

$$l(\beta_0, \beta_1) = \prod_{i:y_i=1} p(x_i) \prod_{i:y_i=0} (1 - p(x_i)),$$

where

$$p(x_i) = \frac{e^{\beta_0+\beta_1 x_i}}{1 + e^{\beta_0+\beta_1 x_i}}$$

Note that the likelihood function $l(\beta_0, \beta_1)$ is the probablity of observing the classes $y_1, ..., y_n$ for correspongding feature values $x_1, ..., x_n$ when the logistic regression model with parameters β_0 and β_1 is used.

The estimates $\widehat{\beta}_0$ and $\widehat{\beta}_1$ are chosen to maximize the likelihood function, i.e., $(\widehat{\beta}_0, \widehat{\beta}_1)$ is a solution to the optimization problem

$$\max_{\beta_0,\beta_1} l(\beta_0, \beta_1).$$

Logistic Regression. Estimating β_0 and β_1

Remark. With $y_i \in \{0, 1\}$, $i = 1, ..., n$, we can write

$$\begin{aligned} l(\beta_0, \beta_1) &= \prod_{i:y_i=1} p(x_i) \prod_{i:y_i=0} (1 - p(x_i)) \\ &= \prod_{i=1}^{n} p(x_i)^{y_i} (1 - p(x_i))^{1-y_i}, \end{aligned}$$

which suggests that it might be useful to take the log of the expression above, leading to the log-likelihood function $\mathcal{L}(\beta_0, \beta_1)$,

$$\begin{aligned} \mathcal{L}(\beta_0, \beta_1) &= \log(l(\beta_0, \beta_1)) \\ &= \sum_{i=1}^{n} \left[y_i \log(p(x_i) + (1 - y_i) \log(1 - p(x_i)) \right]. \end{aligned}$$

Maximizing the likelihood function $l(\beta_0, \beta_1)$ is the same with maximizing the log-likelihood $\mathcal{L}(\beta_0, \beta_1)$, i.e. $\widehat{\beta}_0, \widehat{\beta}_1$ solve

$$\max_{\beta_0, \beta_1} \mathcal{L}(\beta_0, \beta_1).$$

Multiclass logistic regression

We consider the logistic regression model with more than two classes and p ($p \geq 1$) features.

This generalization can be easilly derived from the simple logistic regression model by taking

$$p_k(X) := Pr(Y = k|X) = \frac{e^{\beta_{0k}+\beta_{1k}X_1+\ldots+\beta_{pk}X_p}}{\sum_{j=1}^{K} e^{\beta_{0j}+\beta_{1j}X_1+\ldots+\beta_{pj}X_p}},$$

where $X = (X_1, \ldots, X_p)^T$ and $k = 1, \ldots, K$.

Remark. We note that only $K - 1$ functions are needed since

$$p_K(X) = 1 - p_1(X) - \ldots - p_{K-1}(X).$$

Multiclass logistic regression is also referred to as multinomial (logistic) regression.